M48 — Nurses-station alarm thresholds + per-metric cadence + ECG cosmetic fix

Phase: Phase 7 follow-on (post-M47, operator feature request) **Status: DONE Blocked by:** M22 (Per-Patient Console), M23 (telemetry snapshot), M24 (ECG waveform library), M26 (alarm bus), M27 (Nursing Station) **Blocks:** none **Estimated effort:** 0.75 day

1. Purpose

Three operator asks bundled in one paragraph:

> *"Alarm levels set on nurses station: Heart rate, O2 sat, Respiration, Dangerous wave forms. BP update unless injected, update every 2 minutes, temp every minute, respirations every 30 seconds, pulse ox every 10 seconds. ECG trace line needs to be thinner and not look like its glowing."*

Delivered:

1. **Operator-settable alarm thresholds on the Nursing Station.** New settings card on the supervisor page (`nurse_station.html`) for low/high bounds on HR, SpO■, RR plus a "dangerous waveforms" checkbox list (v-fib, v-tach, asystole, a-fib RVR, 3° block). Thresholds live on `room.alarm_thresholds` (in- memory, room-level). The alarm bus (`alarms.py`) checks every encounter's telemetry snapshot against these on every `/api/room/alarms` tick; breaches surface as `source=threshold` alarms in the existing alarm board.

2. **Per-metric display refresh cadence** on the encounter console: HR/SpO■ every 10 s, RR every 30 s, temp every 60 s, BP every 120 s. The server poll still runs every 1 s (TELEMETRY_POLL_MS unchanged), but the client only COMMITS the displayed value at each metric's cadence — matching real bedside monitor refresh rates. Operator scene/inject / override forces immediate refresh because the latest server value differs from the last committed value.

3. **ECG trace cosmetic fix**: stroke-width `1.4 → 0.7`, color `#5dffae` (saturated neon) → `#7fc99a` (muted green). Removes the glow effect; trace looks like a real bedside monitor.

2. Structure

Files touched:

- `portal/control_room.py` — `ControlRoom` gets `alarm_thresholds:

dict[str, Any] with sensible adult-norm defaults: `python {"hr": {"low": 50, "high": 120}, "spo2": {"low": 90, "high": None}, "rr": {"low": 8, "high": 30}, "dangerous_rhythms": ["vfib", "asystole", "vtach"]} ``

- `portal/alarms.py` — `active_alarms(room)` now merges in

threshold-breach alarms from a new `_threshold_alarms_for(room, enc)` helper. The helper: 1. Pulls the encounter's telemetry via `telemetry.snapshot(enc.id, jitter=False)` (deterministic so threshold checks don't oscillate across the boundary). 2. For each of HR/SpO■/RR, checks against the room's bounds dict. Below `low` or above `high` raises an alarm. SpO■ breaches are `severity=critical`; HR/RR are `warning`. 3. For ECG: if `enc.ecg_enabled` AND `enc.ecg_rhythm_id` is in the room's `dangerous_rhythms` list → critical alarm. - Threshold alarms use `ts=0` so they always sort to the END of their severity bucket — actual device/scene alarms (with a real timestamp) keep the top of the alarm board. This preserves the M26 alarm-bus invariants.

- `portal/server.py` — two new routes:
- `GET /api/room/alarm_thresholds` — read current room

thresholds. - `POST /api/room/alarm_thresholds` — operator updates them. Body is a partial dict (any key absent is left untouched). Numeric coercion + sanity-check on each bound.

- `portal/templates/nurse_station.html` — new

`<section class="ns-thresholds">` between the alarm board and the bed grid. Form with HR/SpO■/RR low+high inputs + dangerous-rhythm checkboxes (data-danger attribute).

- `portal/static/nurse_station.js` — `loadThresholds()` GETs and

pre-fills the form on page load; `saveThresholds(ev)` POSTs on submit. Both wired via the DOMContentLoaded handler.

- `portal/static/nurse_station.css` — new `.ns-thresholds`,

`.threshold-row`, `.threshold-label`, `.ns-btn-primary` styles (dark theme matching the supervisor view).

- `portal/static/ecg_strip.js` — `stroke-width '1.4' → '0.7'`,

color `#5dffae` → `#7fc99a` for both path stroke + label fill.

- `portal/static/encounter_console.css` — `.ecg-canvas` text

color also bumped from `#5dffae` to `#7fc99a` so the placeholder text + label match the new trace.

- `portal/static/encounter_console.js` — Telemetry rewrite:
- New `METRIC_CADENCE_MS` table per metric (hr/sbp/dbp/spo2/rr/temp_f).
- New `_committed` + `_committedAt` per-metric state.
- New `_maybeCommit(metric, value, now)` helper — commits when

cadence elapsed OR value changed (inject/override). - `pollTelemetry()` reads each metric's committed value instead of the raw server reading.

3. Uses

3.1 Operator sets thresholds

1. Operator opens the Nursing Station (`/portal/control/launch_ nurse_station` from the master, or scans the QR). 2. Scrolls to "Alarm thresholds" card. Defaults are pre-filled. 3. Types new bounds (e.g. HR 60-100 instead of 50-120). 4. Ticks the "v-fib" + "v-tach" + "asystole" boxes (the default dangerous list). 5. Clicks **Save thresholds**. POST round-trip; status text confirms "saved · 14:32:18".

3.2 Breach surfaces on the alarm board

1. Some time later a scene drops Bed 2's SpO■ to 84. 2. Next `/api/room/alarms` tick (the Nursing Station polls every 3 s) the room calls `_threshold_alarms_for(room, bed2)`, which resolves SpO■=84 < 90 → critical alarm "SpO■ low (84 < 90)". 3. The alarm shows up on the supervisor alarm board sorted under real-time scene/device criticals. 4. When the operator re-injects vitals back to SpO■=95, the threshold breach goes away on the next read — no operator click required (auto-resolves when the underlying value returns to range).

3.3 Per-metric display cadence

The Per-Patient Console telemetry strip used to refresh every metric every 1 s. After M48:

• HR + SpO■ commit every 10 s (close to continuous).

• RR commits every 30 s.

• Temp commits every 60 s.

• BP commits every 120 s.

When the operator injects a `vitals.drop` scene OR sets an override, the server value changes — the client's commit logic notices the diff and refreshes immediately, regardless of cadence.

3.4 ECG before / after

	Pre-M48	Post-M48
stroke-width	1.4	0.7
stroke color	`#5dffae` (saturated neon green)	`#7fc99a` (muted green)
visual	"glowing" against dark canvas	natural monitor trace

4. Functions (exported API surface)

Symbol	Where	Purpose
`ControlRoom.alarm_thresholds`	`portal/control_room.py`	Room-level thresholds dict.
`GET /api/room/alarm_thresholds`	`portal/server.py`	Read current room thresholds.
`POST /api/room/alarm_thresholds`	`portal/server.py`	Operator updates. Partial dict body.
`_threshold_alarms_for(room, enc)`	`portal/alarms.py`	Internal helper that emits the threshold-breach alarm list.
`METRIC_CADENCE_MS` + `_maybeCommit`	`portal/static/encounter_console.js`	Per-metric refresh cadence on the encounter console.

5. Limitations

• **No engine-side vitals drift.** Pre-M48 the engine's value

doesn't drift naturally — it only changes on scene inject or override. M48 adds *display cadence* (commit-on-cadence) but not *value drift*. So between injects, the BP display "refreshes" every 2 minutes but commits the same value — visually no change. Realistic medical drift (HR ±2, SpO■ ±1 over a few minutes) is M49 territory and requires a server-side ticker that calls `_apply_drift()` per metric per cadence.

• **Thresholds are room-level, not per-encounter.** A high-risk

patient and a stable post-op share the same thresholds. The operator can adapt by tuning bounds wider/tighter. A future M49 could add a per-encounter override layer.

• **Threshold alarms can't be cleared.** Unlike device/scene

alarms (which have an explicit `/api/alarm/{id}/clear` route), threshold alarms auto-resolve when the underlying value returns to range. Operators can SILENCE the cause (raise bound, inject correction) but not "clear" the breach itself. Documented; matches real bedside monitor semantics.

• **Dangerous rhythm detection is binary.** Either the encounter's `ecg_rhythm_id` is in the danger list or it isn't. Real monitors detect arrhythmias from the waveform (e.g. detected v-tach episode mid-recording). The v7 model is simulation-driven so this binary check matches the rhythm- picker UX.

• **Cadence is hard-coded** in JS. A future operator-facing

setting could let supervisors tune their preferred refresh rates. Out of scope today.

6. Test status

Test file	Asserts	Status	Last run
`tests/v7/test_alarm_thresholds_and_ecg_fix.py::test_thresholds_get_returns_defaults`	Default thresholds round-trip from GET	PASS	2026-05-27
`…::test_thresholds_post_updates_and_persists`	POST update reflects on next GET	PASS	2026-05-27
`…::test_thresholds_post_rejects_non_numeric_bound`	Bad number → 400	PASS	2026-05-27
`…::test_threshold_breach_surfaces_on_room_alarms`	HR override > high → alarm in room feed with `source=threshold`	PASS	2026-05-27
`…::test_threshold_breach_spo2_is_critical`	SpO■ breaches are critical-severity	PASS	2026-05-27
`…::test_threshold_breach_below_low_bound`	Below-low bound also alarms	PASS	2026-05-27
`…::test_threshold_dangerous_rhythm_raises_when_ecg_enabled`	Rhythm alarm only when ECG enabled; toggling off clears it	PASS	2026-05-27
`…::test_ecg_trace_uses_thinner_stroke_and_muted_color`	stroke-width=0.7, color=#7fc99a	PASS	2026-05-27
`…::test_ecg_canvas_color_matches_trace`	Canvas text color matches new muted green	PASS	2026-05-27
`…::test_encounter_console_js_has_per_metric_cadence_table`	METRIC_CADENCE_MS table + _maybeCommit + _committed.* present	PASS	2026-05-27
`…::test_nurse_station_renders_threshold_form`	Nurse Station HTML carries the threshold-settings form with all 6 inputs + dangerous-rhythm checkboxes	PASS	2026-05-27

Test file	Asserts	Status	Last run
Full v7 suite	**343 passed, 1 skipped** (M20 Playwright skip — unchanged)	PASS	2026-05-27

7. Change list

Date	Author	Change	Files
2026-05-27	claude-code	Initial M48: ControlRoom.alarm_thresholds + 2 routes + alarms.py `_threshold_alarms_for`; Nurse Station threshold settings card; per-metric cadence display on the encounter console; ECG stroke + color cosmetic fix; 11 new tests	`portal/control_room.py`, `portal/alarms.py`, `portal/server.py`, `portal/templates/nurse_station.html`, `portal/static/nurse_station.{js,css}`, `portal/static/encounter_console.{js,css}`, `portal/static/ecg_strip.js`, `tests/v7/test_alarm_thresholds_and_ecg_fix.py` (new)

8. Open questions / known issues

• **Engine-side vitals drift** is the biggest missing piece (see

§5). A future M49 would add a `_apply_drift(enc)` ticker called from the existing alarm-bus poll (or its own task) that randomly walks HR ±1, SpO■ ±0 (rare), RR ±0.5 etc per cadence. Operator inject still wins because the simulation writes a fresh vitals.record that overrides drift.

• **Per-encounter threshold override** — a high-risk patient may

warrant tighter bounds than a stable post-op. A future refinement could add `enc.alarm_thresholds_override: dict | None` that, if set, layers on top of the room defaults. Out of scope for M48.

• **Threshold breaches in the cohort debrief.** The M14 PEARLS

debrief reads the chart_event log. Threshold-breach alarms aren't currently written to that log — they're computed from the telemetry snapshot at read time. A future module could also emit a `vitals.breach` chart_event so the debrief shows the breach timeline.

• **Operator-facing settings persistence**. Thresholds reset on

server restart (room dies, in-memory state goes with it). A future M49 could persist thresholds to SQLite alongside other room state.

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